Word maps close to the identity

Terence Tao · 5 February 2019

https://terrytao.wordpress.com/2019/02/05/word-maps-close-to-the-identity/

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While talking mathematics with a postdoc here at UCLA (March Boedihardjo) we came across the following matrix problem which we managed to solve, but the proof was cute and the process of discovering it was fun, so I thought I would present the problem here as a puzzle without revealing the solution for now.

The problem involves word maps on a matrix group, which for sake of discussion we will take to be the special orthogonal group $SO(3)$ of real 3×3 matrices (one of the smallest matrix groups that contains a copy of the free group, which incidentally is the key observation powering the Banach-Tarski paradox). Given any abstract word w of two generators x, y and their inverses (i.e., an element of the free group $\mathbf{F}_2$), one can define the word map $w : SO(3) \times SO(3) \to SO(3)$ simply by substituting a pair of matrices in $SO(3)$ into these generators. For instance, if one has the word $w = xyx^{-2}y^2x$, then the corresponding word map $w : SO(3) \times SO(3) \to SO(3)$ is given by

$$w(A, B) := ABA^{-2}B^2A$$

for $A, B \in SO(3)$. Because $SO(3)$ contains a copy of the free group, we see the word map is non-trivial (not equal to the identity) if and only if the word itself is nontrivial.

Anyway, here is the problem:

> **Problem.** Does there exist a sequence $w_1, w_2, \ldots$ of non-trivial word maps $w_n : SO(3) \times SO(3) \to SO(3)$ that converge uniformly to the identity map?

To put it another way, given any $\varepsilon > 0$, does there exist a non-trivial word w such that $\|w(A, B) - 1\| \leq \varepsilon$ for all $A, B \in SO(3)$, where $\| \|$ denotes (say) the operator norm, and 1 denotes the identity matrix in $SO(3)$?

As I said, I don't want to spoil the fun of working out this problem, so I will leave it as a challenge. Readers are welcome to share their thoughts, partial solutions, or full solutions in the comments below.

Comments (14)

Anonymous · 5 February 2019

Professor Tao, Happy Lunar New Year!

Jie · 5 February 2019

Prof Tao, happy lunar new year!

Andreas Thom · 5 February 2019

I proved this and more general results in "Convergent sequences in discrete groups", Canad. Math. Bull. 56 (2013), no. 2, 424-433. The preprint is available at https://arxiv.org/abs/1003.4093.

Andreas Thom · 5 February 2019

See also https://mathoverflow.net/questions/229384/solving-equations-in-so3-an-open-problem-by-jan-mycielski/229762

Andreas Thom · 5 February 2019

It is an interesting open problem (as far as I can tell) to determine how fast (in terms of the word length) the image of the word map can converge to the identity matrix. This is related to the shortest length of iterated commutators, a problem that was discussed in

Abdelrhman Elkasapy and Andreas Thom, On the length of the shortest non-trivial element in the derived and the lower central series. J. Group Theory 18 (2015), no. 5, 793–804.

Theorem 5.2. discusses the best asymptotics that we could find. See https://arxiv.org/abs/1311.0138 for the preprint.

duetosymmetry · 5 February 2019

Suppose we consider w_i(e,B) where e is the identity element in SO(3). Then w_i(e,B) can only be of the form B^k for some integer k.

Is it then a necessary condition that there is some sequence of k_i's such that B^{k_i} converges to the identity for all B in SO(3)?

> **Sean Eberhard** · 5 February 2019
>
> @duetosymmetry No, because k might be (and will be) zero.

Anonymous · 8 February 2019

https://mathoverflow.net/questions/45483/word-maps-on-compact-lie-groups

Dmitriy Zakharov · 11 February 2019

For $A \in SO(3)$ denote $\text{rot } A = \sup \frac{||Ax-x||}{||x||}$. It is clear that if $\text{rot } A$ is small enough then A is close to the identity.

Lemma. For any $A, B \in SO(3)$ we have $\text{rot } [A,B] \leqslant 2\text{rot } A \cdot \text{rot } B$.
Proof:

$$||[A,B]x - x|| = \left|\left|ABA^{-1}B^{-1}x - x\right|\right|$$

$$= ||ABx - BAx|| \leqslant ||ABx - (Ax + Bx - x)|| + ||BAx - (Ax + Bx - x)||$$

$$\leqslant 2\text{rot } A \text{rot } B\|x\|.$$

Now notice that one of the operators $A, A^2, ..., A^{10}$ has rot at most $1/3$ for any $A \in SO(3)$. So, there is $i \in \{1, ..., 10\}$ so that the n-fold commutator $C_i = [A^i, [A^i, ...[A^i, B]...]]$ has rot at most $2(2/3)^n$. Therefore, the word $[...[C_1, C_2], C_3]...C_{10}]$ has rot at most $2^{10}(2/3)^n$ and it is obviously non-trivial.

Jean Francois LEON · 11 February 2019

I am sorry but I don't understand you lemma.
You establish a bound on the norm of the commuting not its rot.
Do I miss something obvious?

[Fixed some typos I spotted in the proof of that lemma. -T]

stunient · 17 February 2019

How does one see that the word is nontrivial?

jair2018 · 13 February 2019

Reblogged this on Crap.

Anonymous · 13 February 2019

Sorry for the rude blog post,I was experimenting with something on my blog,I called it crap to just label it as a pastebin

Henry · 18 February 2019

I believe we can find a sequence of rotations of S^1 that converges uniformly to the identity map. Then, using the fact that the group of sphere rotation is isomorphic to $SO(3)$, we can define the elements of our sequence by the maps $(A,B) \mapsto T_n B^{-1} B$, where T_n is the nth element of the uniformly convergent sequence from the circle case.